

Assignment 2: Call frames

Joon Lee Richard Li
jl4875 rl998

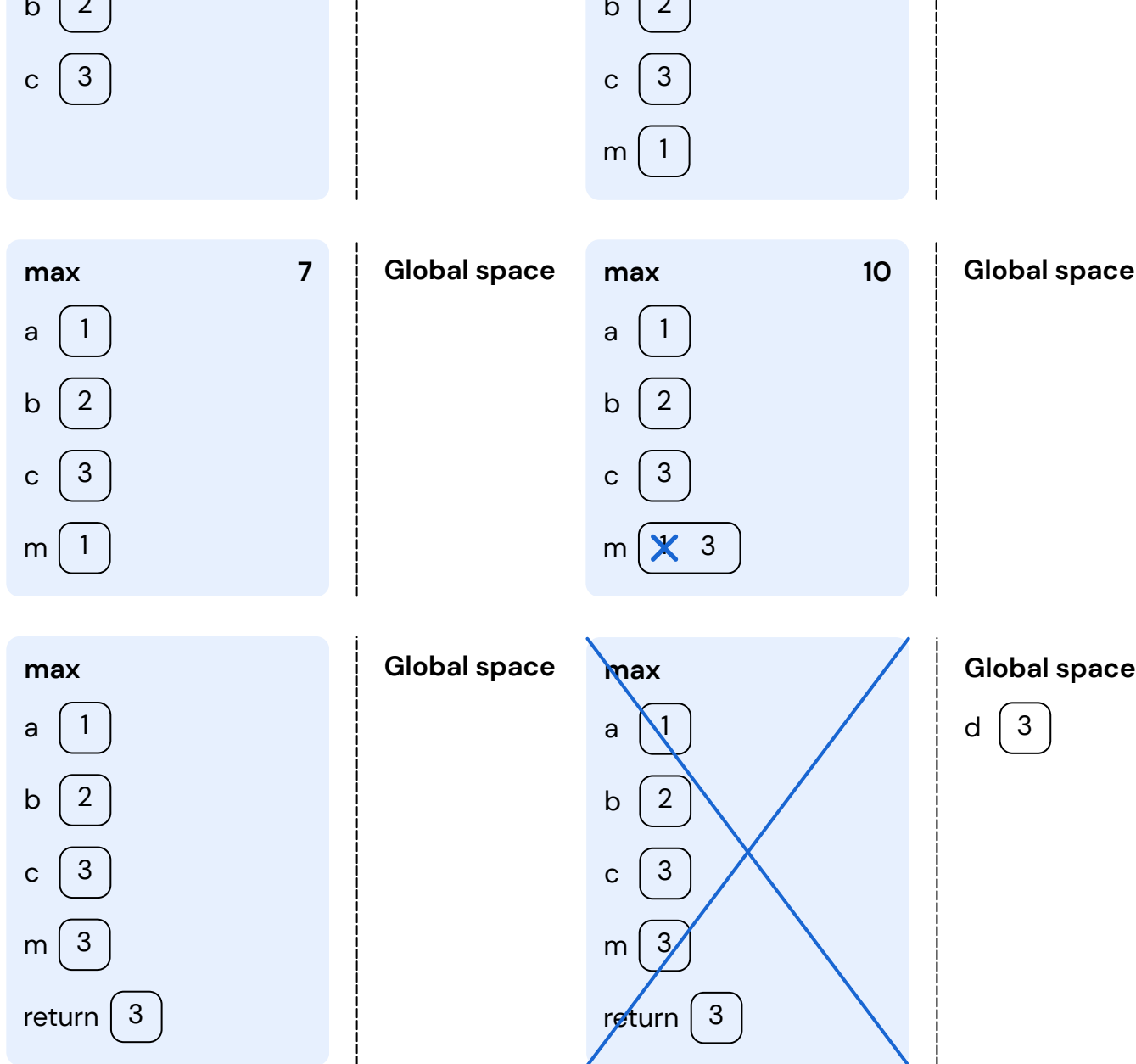
Authors: Joon Lee (jl4875) and Richard Li (rl998)
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Acknowledgments: For Parts A–E, we used the Cornell CS 1110 Python Tutor to check the execution sequences and verify changes to call frames, global space, and the heap.

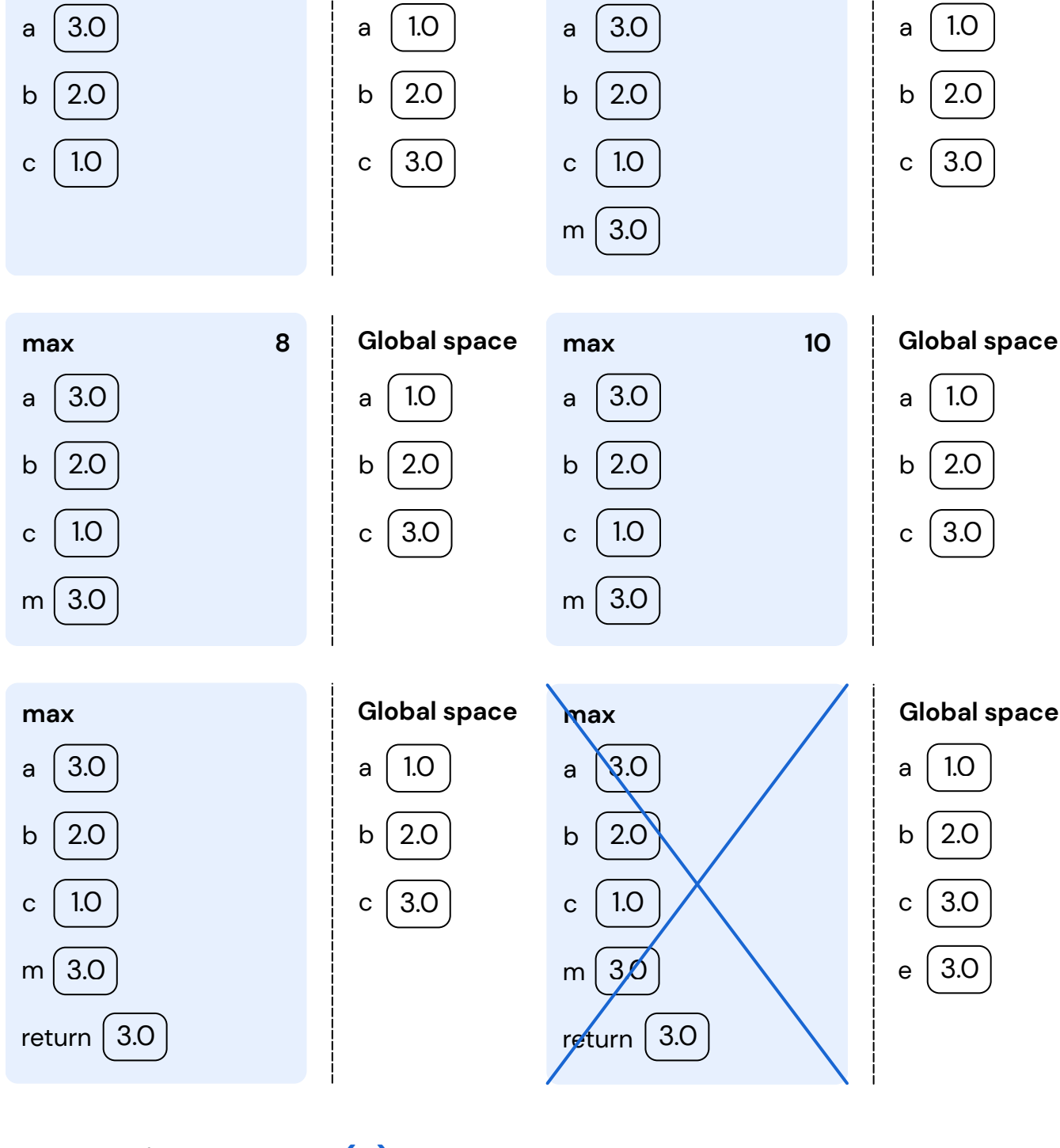
```
1 def max(a,b,c):
2     """Returns: maximum of a, b, and c
3
4     Precondition: a, b, c are numbers"""
5     m = a
6     if (c > b and c > a):
7         m = c
8     elif b > a:
9         m = b
10    return m
```

The function definition for max is reproduced from the instructions for assignment 2.

Part A: Diagram $d = \text{max}(1,2,3)$

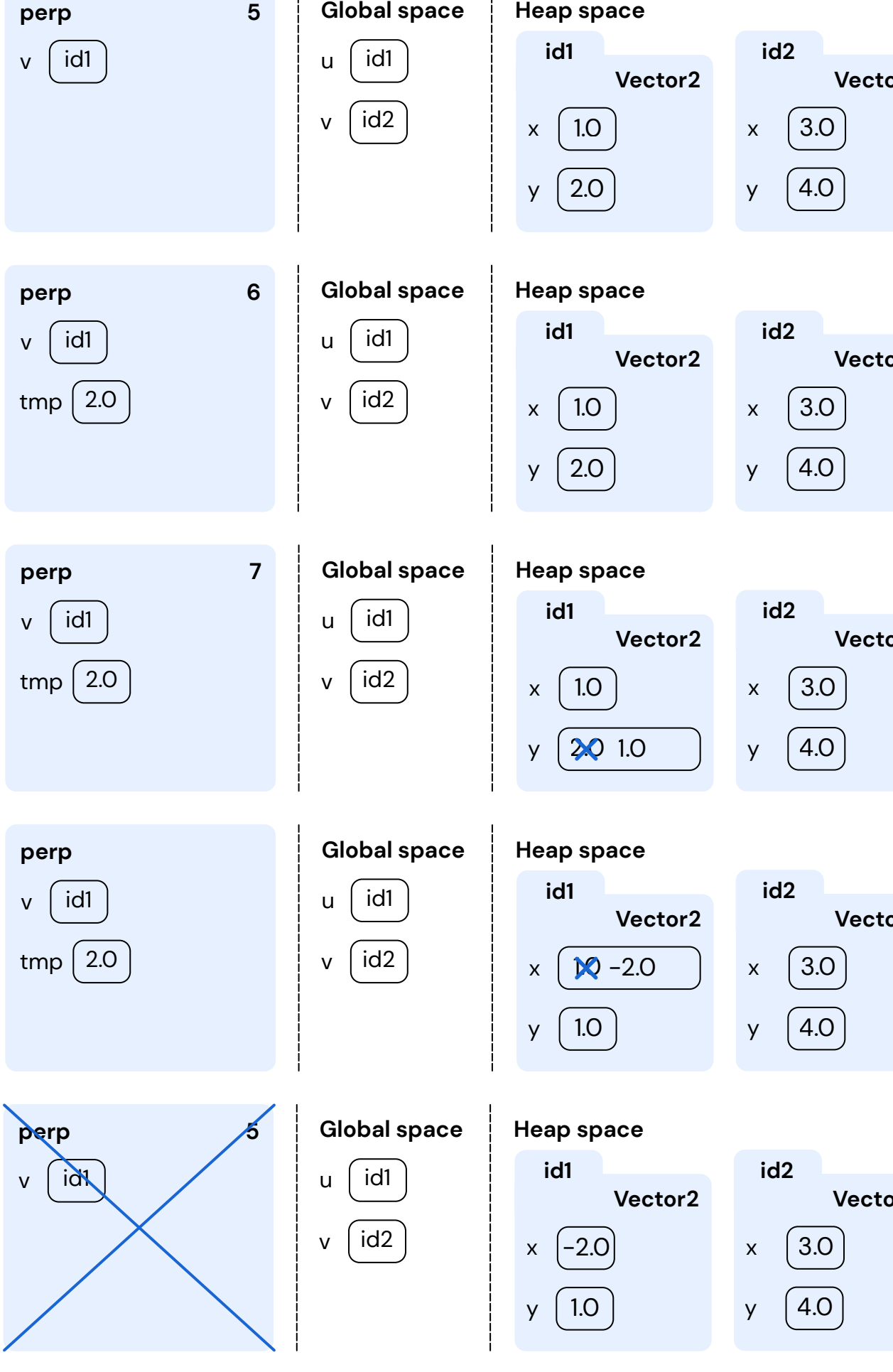


Part B: Diagram $e = \text{max}(c,b,a)$



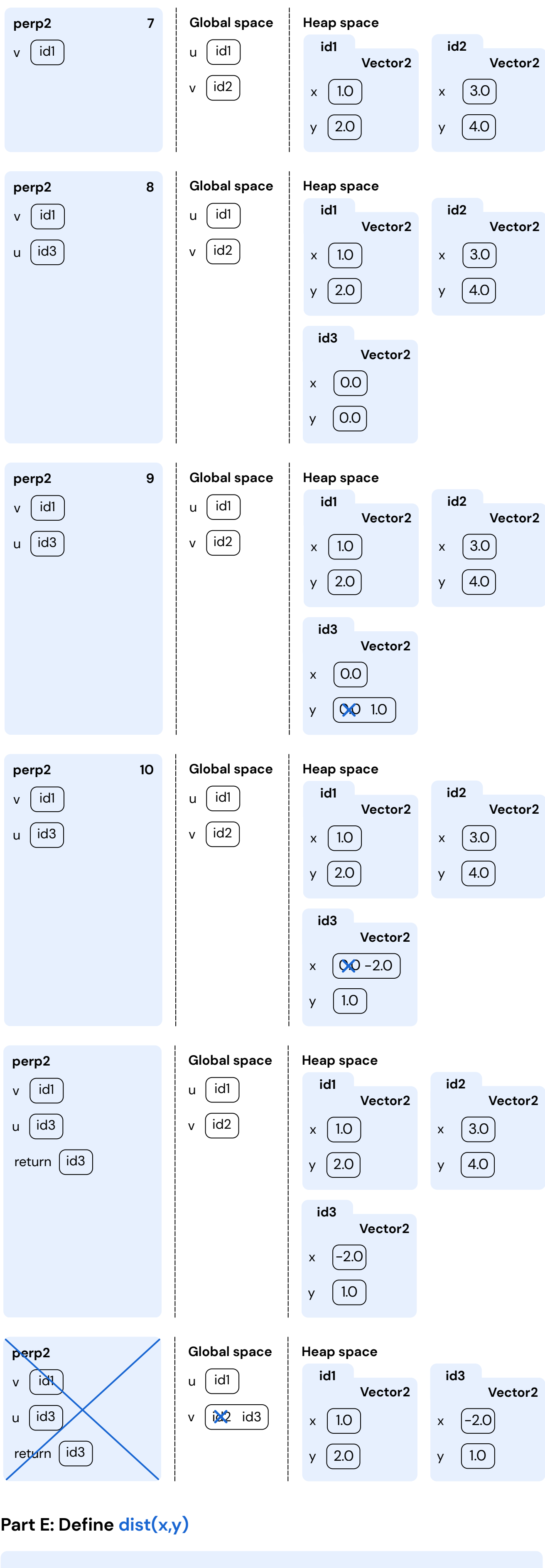
Part C: Diagram $\text{perp}(u)$

```
1 def perp(v):
2     """Modifies v to be perpendicular to the original
3
4     Precondition: v is a Vector2 object"""
5     tmp = v.y
6     v.y = v.x
7     v.x = -tmp
```



Part D: Diagram $v = \text{perp2}(u)$

```
1 import intros
2
3 def perp2(v):
4     """Returns a vector perpendicular to v
5
6     Precondition: v is a Vector2 object"""
7     u = intros.Vector2(0.0,0.0)
8     u.y = v.x
9     u.x = -v.y
10    return u
```



Part E: Define $\text{dist}(x,y)$

```
1 def dist(x,y):
2     """Returns: The number line distance between x and y
3
4     Example: dist(2,5) returns 3
5     Example: dist(5,2) returns 3
6
7     Parameter x: the starting point
8     Precondition: x is a number
9
10    Parameter y: the ending point
11    Precondition: y is a number"""
12    a = x-y
13    if a < 0:
14        b = -a
15    else:
16        b = a
17    return b
```